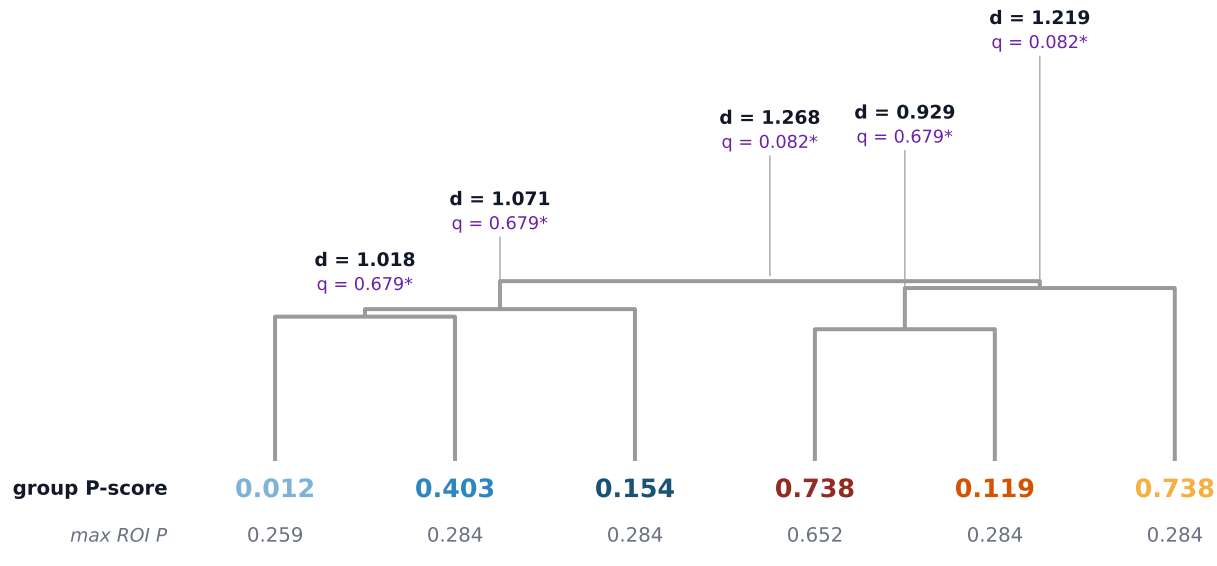
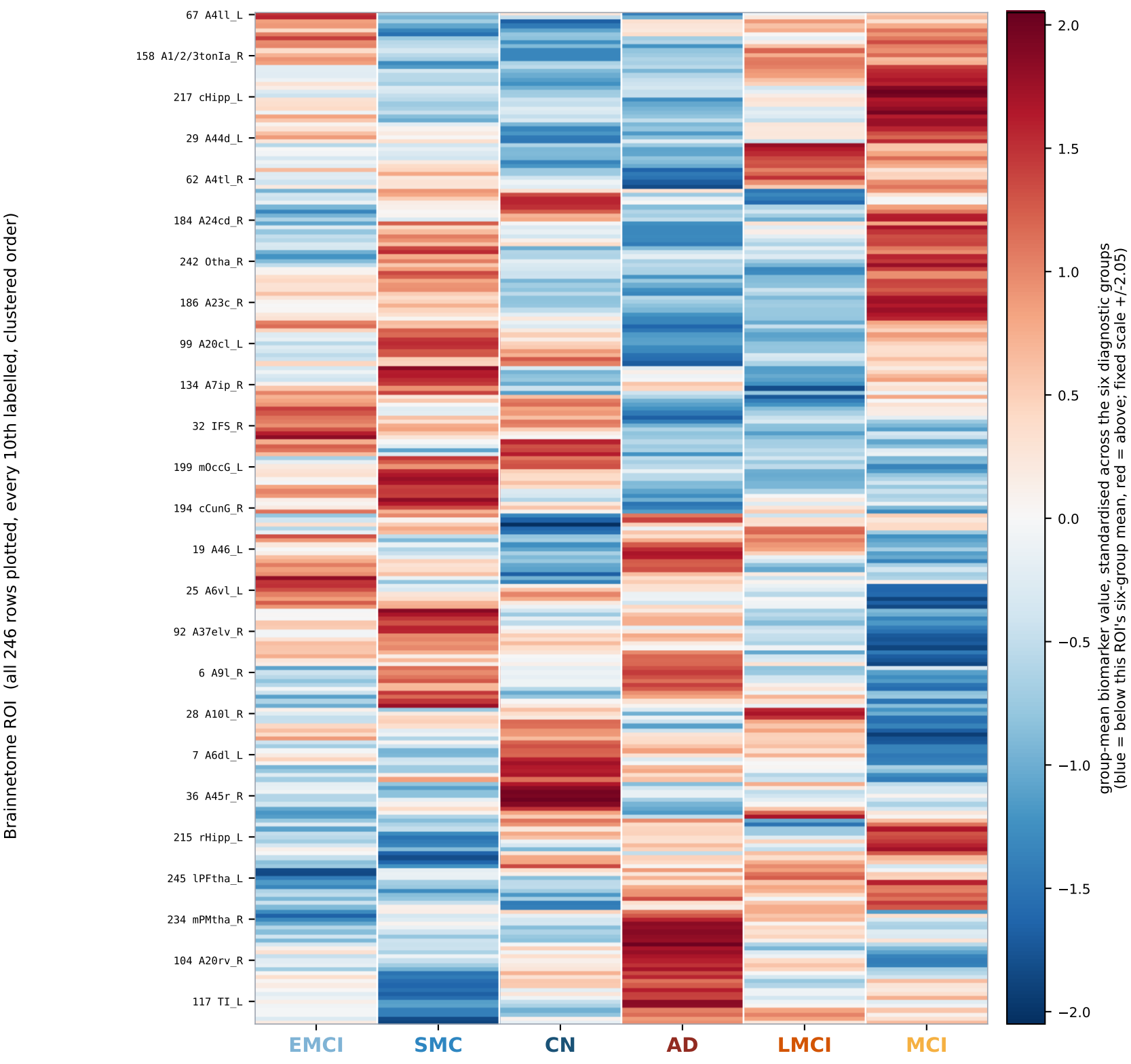
Hierarchical Clustering of Alzheimer's Diagnostic Groups
Regional amplitude of low-frequency fluctuation (mALFF)

cells are the MEAN mALFF of the subjects in each diagnostic group (n = 20/20/20/16/20/20), standardised across the six groups per ROI (ddof = 1)
groups clustered by correlation distance + average linkage over the 246-D profile | ROI rows ORDERED by euclidean + average linkage (tree not drawn)





clinical progression order: CN → SMC → EMCI → MCI → LMCI → AD

bootstrap stability of the branches (1000 resamples of subjects within groups): {LMCI,AD} 46%; {CN,SMC,EMCI} 25%; {SMC,EMCI} 25%

group P-score = -log10(q) from the MULTIVARIATE stage-vs-rest permutation test (24-test family); max ROI P = largest ROI-level P-score, DESCRIPTIVE only

d = merge DISTANCE (geometric, not a p-value). *q = permutation q for that cluster contrast, SELECTIVE (clusters came from these data) -- anti-conservative, descriptive only.

Heatmap shows BIOMARKER ACTIVITY; the top rows show STATISTICAL SIGNIFICANCE. Different quantities. The tree clusters on biomarker activity.

Descriptive only. Branch heights are distances, not p-values. No group reaches q < 0.05 and no ROI x stage test in this project does either.